

Relational Agency

Vrije Universiteit Brussel



Relational agency

Heylighen, Francis

Published in:
Evolution 'On Purpose'

DOI:
[10.7551/mitpress/14642.003.0008](https://doi.org/10.7551/mitpress/14642.003.0008)

Publication date:
2023

Document Version:
Accepted author manuscript

[Link to publication](#)

Citation for published version (APA):
Heylighen, F. (2023). Relational agency: a new ontology for co-evolving systems. In P. Corning (Ed.), *Evolution 'On Purpose': Teleonomy in Living Systems* (pp. 79-104). (Vienna Series in Theoretical Biology). MIT Press.
<https://doi.org/10.7551/mitpress/14642.003.0008>

Differences Between Objects and Agencies

Objects	Agencies
Sensitivity: inert	Sensitivity: sensitive
Activity: passive	Activity: active
Dynamics: mechanical	Dynamics: goal-directed
Shape: rigid	Shape: variable
Positioning: localized	Positioning: localized or distributed
Constituents: material particles	Constituents: processes
Existence: independent	Existence: relational

Key Distinction Between Objects and Agencies:

- The table highlights a fundamental contrast between how we conceptualize inert, passive, and rigid entities (**objects**) versus active, goal-oriented, and relational entities (**agencies**).
- Objects are viewed as static and localized physical entities, while agencies are seen as dynamic, relational, and potentially distributed phenomena.
- By reframing existence as **relational** rather than **independent**, the table and text challenge mechanistic and reductionist views of the world.
- It suggests a paradigm where even entities traditionally viewed as lifeless (like rocks or rivers) can have agency when understood within relational contexts.

....

It is this context-independence of written descriptions that allowed knowledge to be recorded for the long-term and to be disseminated widely without losing its meaning (Heylighen 1999). Thus, for example, ancient Greek treatises about astronomy, geometry and philosophy could still be read and understood two thousand years later in the very different context of the European Renaissance, providing an impetus for the scientific revolution. The further development of mathematical formalisms, such as Cartesian coordinates, calculus and algebra, helped scientists to describe the properties of objects increasingly precisely and

objectively, making abstraction of the observing subject and the context of observation. The resulting formalization allowed knowledge to be expressed in the form of universal “laws of nature” that are independent of space, time or observer, thus forming the foundation of the classical scientific worldview (Heylighen 2012). Such precise, context-independent description moreover facilitated the development and spread of technology, because it allowed engineers to write down a full specification of the components and mechanisms of the objects they designed. Yet, even before formal descriptions, the object-based way of thinking had been stimulated by our use of an ever-larger array of artifacts, such as furniture, pottery or tools. Indeed, unlike natural systems, artifacts are designed to be mere instruments, i.e. passive objects that can be dependably manipulated by their human users— without deploying any agency of their own. You would not want your table to move to a different room because it prefers the view there! Neither would you want your hammer to decide it is better to swing sideways when you want it to swing downwards... As people spent more and more time inside artificial environments, such as houses, cities and offices, their interactions with the environment were increasingly restricted to the perception and handling of such passive objects.

...

Thus, it seems as if the development of civilization, and especially of the symbolic systems supporting science and technology, has alienated humans not just from their natural environment, but from the animistic awareness that makes us intuitively understand and feel part of that living nature (Charlton 2007). Abram and others (Abram 2010; Harvey 2019) have argued that our actions have severely upset the ecological balance—exhausting resources, polluting rivers, and driving many species to extinction—in large part because we see nature as a mere collection of objects to be manipulated, rather than as a network of agencies in which we are intimately involved by a variety of relationships. A renaissance of relational agency? Is there a way to recover the relational understanding needed for a deep ecological consciousness, yet without losing the benefits of objective, scientific description? That would require a conceptual framework and formal language able to accurately represent agencies and their relations, without reducing them to independent objects.

...

The ontology of actions

The ontology of objects assumes that there are elementary objects, called “particles”, out of which all more complex objects—and therefore the whole of reality—are constituted. Similarly, the ontology of relational agency assumes that there are elementary processes, which I will call actions or reactions, that form the basic constituents of reality (Heylighen 2011; Heylighen and Beigi 2018; Turchin 1993). A rationale for the primacy of processes over matter can be found in quantum field theory (Bickhard 2011;

Kuhlmann 2000). Quantum mechanics already made clear that observing some phenomenon, such as the position of a particle, is an action that necessarily affects the phenomenon being observed: no observation without interaction. Moreover, in general the result of that observation is indeterminate before the observation is made: the action of observing in a real sense creates the property being observed, through a process known as the collapse of the wave function (Heylighen 2019; Tumulka 2006). For example, before the observation a particle such as an electron typically does not have a precise position in space, but immediately afterwards it does. More generally, quantum mechanics tells us that microscopic objects such as particles in general do not have objective, determinate properties, but that such properties are (temporarily) generated through interaction (Barad 2003). Quantum field theory adds that the objects (particles) themselves do not have a permanent existence, but that they too can be created or destroyed through interactions, such as nuclear reactions. Particles can even be generated by fluctuations of the vacuum (i.e. out of nothing), albeit that in this case their existence is so transient that they are called “virtual” (Milonni 2013). At a larger scale, the molecules that constitute living organisms are similarly ephemeral, being both produced and broken down by the chemical reactions that constitute the organism’s metabolism. Cells and organelles in the body too are in a constant flux, being broken down by processes such as apoptosis and autophagy, while new ones are grown through cell division and from stem cells. Similar creation- destruction processes can be found at the level of ecosystems, where relations of predation, symbiosis, and reproduction between organisms join with meteorological and geological forces to produce a constantly changing landscape of forests and rivers, mountains and meadows. Even the planet is in a constant flux, as magma moves up, down and sideways under its mantle, moving continents and opening up volcano-studded rifts. And the sun and stars that irradiate it are merely boiling cauldrons of nuclear reactions generating new elements in their core, while releasing immense amounts of energy.

...

In this notation, the condition-action rules can be interpreted as reactions, in the sense that the conditions {a, b, ...} on the left-hand side of the arrow “react” with each other in order to produce the novel conditions {e, f, ...} on the right-hand side of the arrow. Such a reaction represents an elementary process that takes the elements on the left as its inputs and transforms them into the elements on the right as its outputs (Heylighen, Beigi, and Veloz 2015). Agencies (A) can now be defined in this framework as necessary conditions for the occurrence of a reaction, which however are not themselves affected by the reaction:



...

The reaction between A, X and Y can therefore be reinterpreted as an action performed by the agency A on condition X in order to produce condition Y. This can be represented in shorter notation as: $A: X \rightarrow Y$ Note that while an agency remains invariant during

the reactions it catalyzes, there in general exist other reactions that destroy (consume) or create (produce) that agency. Thus, while agencies can play the role reserved in the old ontology for objects or forces, they are neither inert nor invariant. An agency will in general catalyze several reactions. This means that it will react to different conditions by different actions so as to produce different new conditions. For example: $A: X \rightarrow Y, Y \rightarrow Z, U \rightarrow Z$ This set of actions triggered by A can be interpreted as a dynamical system that maps initial states (X, Y, U) onto subsequent states (Y, Z, Z) (Heylighen 2022; Sternberg 2010).

This paper presents a comprehensive exploration of Delta B (ΔB), a quantitative measure of structural deviation from an ideal state of stasis, using the recently discovered Einstein tile as a primary model. By integrating entropy (ΔS) as a fundamental descriptor of disorder, we propose a mathematical framework that characterizes deviations from perfect aperiodicity. We expand upon the concept of stasis-not merely as a physical state but as a multifaceted equilibrium encompassing mental, emotional, psychological, physiological, intellectual, and philosophical dimensions. Due to the inherent complexity of harmonizing these six aspects (the "six-body problem"), true stasis remains unattainable, yet it represents a universal pursuit in both physical systems and life forms. Through the definition of a quadratic function derived from thermodynamic principles, we analyze the behavior of ΔB in relation to entropy changes. This provides new insights into the stability and transformations of systems striving for stasis under varying conditions of order and disorder.

The next step in the development of the object-based worldview, due to thinkers such as Galileo, Descartes and Newton, was the insight that properties can be measured, i.e. determined in a precise, objective manner as quantities. The essential properties remain invariant, but the contingent ones, such as temperature, position or velocity, can vary over the course of time. The corresponding variables define a state space or phase space. Changes in the values of these variables are described by dynamical equations, which express the effect of different forces, such as gravitation, on the objects. Thus, all change can be reduced to the deterministic, time- parameterized trajectory of a collection of invariant objects through their state space (Heylighen, Cilliers, and Gershenson 2007).

...

The ontology of objects assumes that all phenomena we observe can be reduced to objects and their movement in space. Objects are separate, invariant pieces composed of some inert substance, which we call matter. Matter may be repositioned, divided or reshaped, but it cannot be destroyed or created. It just stays there, passively waiting until some force may move it into a different spatial arrangement. By assembling elementary pieces of matter, which we call particles, into different forms or shapes, different objects can be created. But the objects are intrinsically just as passive, inert or insensitive as the matter that constitutes them. Unless some force acts on them, they remain as they are: rigid, invariant, unchanging. Objects are

distinguished by their location in space. No two objects can occupy the same space: some minimal distance must separate them. Therefore, distinct objects are intrinsically independent: you can move the one without affecting the others. Objects can be identified by their form, i.e. the geometry of the matter that constitutes them. Form and position determine their properties. Therefore, properties are intrinsic, absolute and objective. They do not depend on the presence or absence of other objects, including those constituting what we call an observer, environment, or context.

...

Complex objects (“systems”) that are constituted of smaller objects (“parts”) inherit their properties from these parts. There are no “emergent” properties: the whole does not have anything that is more than the sum of its parts. Therefore, to understand the behavior of the whole, it is sufficient to accurately observe the changes in the properties of its parts, i.e. in the variables that describe the system. These changes are driven by external forces obeying natural laws. For example, the fundamental object of neo-Darwinist theory is a population. Its variable properties are the frequencies of different genes. Changes in these frequencies are driven by the force of natural selection, here conceived as adaptation to a given, external environment.

In earlier work (Heylighen and Beigi 2018), I have proposed a radical resolution to this conundrum, by interpreting both objects and subjects as special cases of the more general category of agents, i.e. phenomena that act on other phenomena. The remainder of this text will spell out some of the implications of this ontology for understanding biological organization and its evolution, by interpreting not only organisms, but their physical components as well as the ecosystems they collectively form, as interrelated agencies. Table 1 provides a first sketch of how agencies differ from objects, a summary that will be elaborated in the rest of the paper.

`'Relational agency: Heylighen, Francis(2023)' abstracted`

The Ontology of Actions

The ontology of objects assumes that there are elementary objects, called “particles,” out of which all more complex objects—and therefore the whole of reality—are constituted. Similarly, the ontology of relational agency assumes that there are elementary processes, which I will call ****actions**** or ****reactions****, that form the basic constituents of reality (Heylighen 2011; Heylighen and Beigi 2018; Turchin 1993).

A rationale for the primacy of processes over matter can be found in **quantum field theory** (Bickhard 2011; Kuhlmann 2000). Quantum mechanics has shown that observing some phenomenon, such as the position of a particle, is an action that necessarily affects the phenomenon being observed: **no observation without interaction**. Moreover, the result of that observation is often indeterminate before the observation is made. The action of observing, in a real sense, creates the property being observed through a process known as the **collapse of the wave function** (Heylighen 2019; Tumulka 2006).

For example:

- Before observation, a particle (e.g., an electron) typically does not have a precise position in space.
- Immediately after observation, the particle assumes a precise position.

More generally, quantum mechanics tells us that:

- Microscopic objects, such as particles, do not have objective, determinate properties.
- Such properties are (temporarily) generated through interaction (Barad 2003).

Quantum field theory expands on this, asserting that:

- **Objects (particles)** themselves do not have permanent existence.
- They can be created or destroyed through interactions, such as nuclear reactions.
- Particles can even be generated by **vacuum fluctuations** (Milonni 2013), though such particles are so transient that they are called “virtual.”

Processes in Living Organisms and Ecosystems

At larger scales:

- Molecules in living organisms are ephemeral, produced and broken down by the chemical reactions of metabolism.
- Cells and organelles are in constant flux, undergoing processes like **apoptosis** and **autophagy**, while new cells are formed through **cell division** and **stem cell differentiation**.

In ecosystems:

- Processes such as **predation**, **symbiosis**, and **reproduction** interact with **meteorological** and **geological forces** to produce constantly changing landscapes of forests, rivers, mountains, and meadows.

Even at planetary and cosmic scales:

- The Earth's crust and mantle are in flux, with magma moving continents and forming volcanoes.
- The Sun and stars are boiling cauldrons of nuclear reactions, generating new elements in their cores while releasing immense amounts of energy.

Actions, Reactions, and Agencies

In this framework:

- **Condition-action rules** can be interpreted as reactions:

$\{a, b, \dots\} \rightarrow \{e, f, \dots\}$

This represents an **elementary process** where:

- The conditions on the left ($\{a, b, \dots\}$) act as inputs.
- These inputs transform into the conditions on the right ($\{e, f, \dots\}$), which are the outputs (Heylighen, Beigi, and Veloz 2015).

Definition of Agency

Agencies (**A**) can be defined as **necessary conditions** for the occurrence of a reaction. However, agencies themselves are not directly affected by the reaction:

$A + X \rightarrow A + Y$

Here:

- The reaction between **A**, **X**, and **Y** can be reinterpreted as an **action** performed by agency **A** on condition **X** to produce condition **Y**.
- This can be represented in shorter notation as:

$A: X \rightarrow Y$

Dynamic Properties of Agencies

While an agency remains invariant during the reactions it catalyzes:

- There exist reactions that ****create**** (produce) or ****destroy**** (consume) that agency.

Thus, agencies are:

- Neither inert nor invariant.
- They catalyze multiple reactions and respond dynamically to different conditions:

$\text{'A: } X \rightarrow Y, Y \rightarrow Z, U \rightarrow Z \text{'}$

This set of actions triggered by ****A**** can be interpreted as a ****dynamical system****, mapping initial states (e.g., X, Y, U) onto subsequent states (e.g., Y, Z, Z) (Heylighen 2022; Sternberg 2010).

Dynamical Systems and Attractors

A dynamical system typically has one or more **attractors**—these are states or sets of states to which the system's dynamics converges. Processes catalyzed by the agency (A) will lead into the attractor, but (A) does not provide a way out. Once in the attractor, further actions can only produce states that are also within the attractor.

An attractor is surrounded by a **basin of attraction**, which contains the states that converge to the attractor but are not part of it. For example, in a system where (Z) is the attractor, states like (X), (Y), and (U) may lie in its basin. Processes starting from these conditions inevitably end in (Z), with no possibility of return.

This attractor-centered dynamics models **goal-directedness** or **teleonomy** (Heylighen 2022; Heylighen & Beigi 2018). The actions executed by agency (A) aim towards the goal (Z), which serves as the shared final state or "end" of processes starting from different initial states ((X), (Y), (U))—a feature called **equifinality** (Lyman 2004) or **plasticity**.

Perturbations—external processes not controlled by (A)—that deviate a state from its goal-directed trajectory will generally be counteracted by (A), ensuring the state still ends up in the attractor (as long as the deviation remains within the basin). This characteristic is called **persistence** or **regulation through negative feedback** (Heylighen 2022).

Thus, an agency enables a bundle of actions that act on the present situation to drive it toward a restricted set of attractor states. These attractor states define the implicit **goal**, "preference," or **end** of the agency.

Examples of Agency and Attractors

1. Bacterial Chemotaxis

Bacteria exhibit **chemotaxis**, where they seek food by swimming up gradients of food molecules toward the source (Sourjik & Wingreen 2012). In this case:

- The source (highest food concentration) functions as the attractor of the chemotaxis dynamics.
- The bacterial movement follows a simple **condition-action rule**:
If the concentration of food molecules does not increase, change direction.

The result: The bacterium persistently moves toward regions of increasing food concentration until it reaches the attractor (highest concentration). Here, the **agent** is the bacterium, a **spatially bounded system** driving food-directed movement.

2. Gravity as an Agency

Gravity exemplifies an agency with no clear localization. It catalyzes the reaction:

[
 $\text{\text{unsupported heavy object}} + \text{\text{gravity}} \rightarrow \text{\text{falling object}} + \text{\text{gravity}}$
]

The attractor in this dynamic is the lowest possible position the object can reach until it is supported again. We may say gravity "prefers" lowering the positions of heavy objects.

3. Thunderstorms as Agencies

A thunderstorm is an example of a **non-localized agency**. Its actions include producing rain, wind, and lightning, with the implicit "goal" of discharging its water onto the earth.

- Historically, thunderstorms were attributed to gods like Thor or Zeus.
- Today, we analyze them via complex meteorological models.

Regardless of interpretation, the thunderstorm's impact on its surroundings is significant. Understanding its **typical behavior** allows us to anticipate and adapt to its actions. For example:

- People outside during a thunderstorm will get wet.
- Tall structures may be struck by lightning.
- Rivers may flood.

An **object-based ontology** provides little insight into thunderstorms. A thunderstorm is not an object in any practical sense, and reducing it to the water and air molecules it contains offers no meaningful understanding of its behavior.

...

That course of action is in general unpredictable because the environment will constantly throw up challenges that demand some novel action from the agent, so as to remain on course to the goal (Heylighen 2012). The reason is that the environment consists of other agencies. These similarly react to challenges by actions that produce new conditions. These in turn may challenge one or more agent to act.

...

Relations between actions and agencies Unlike objects, the reactions that form the elements of our new ontology are intrinsically coupled. That is because the conditions that constitute the inputs and outputs of a reaction always must occur in some other reaction as well. This is the principle of “the difference that makes a difference” (Bateson 2000): for some condition to be distinguishable it must incite a reaction that itself leads to a distinguishable condition. A condition that does not produce any distinguishable effect simply cannot be observed, neither directly nor indirectly. Therefore, we may as well assume that it does not exist. For example, it is in principle possible to reintroduce inert objects in the action ontology by means of the reaction $a \rightarrow a$ (the object a is always followed by the object a , without any change). But if we want to observe whether object a is present, we will need at least a reaction of the form: $a + \text{observation} \rightarrow a + \text{observation result}$. For example, the observation may consist in shining light on the object, while the observation result consists in the detection of light reflected back by the object.

Self-Maintaining Systems

Consider a special type of aquarium that does not exchange any matter with the outside world. Such a hermetically sealed, transparent bowl, called an **"ecosphere"**, contains air, seawater, shrimps, algae, and bacteria (Schilthuizen 2008). The organisms in such a bowl can survive for years without requiring any outside intervention. That is because their activities form a self-maintaining whole that can be summarized by the following reactions:

```
""
→ light
shrimps + algae + O2 → shrimps + waste + CO2 + heat
bacteria + waste + O2 → bacteria + nutrients + CO2 + heat
algae + light + CO2 + nutrients → 2 algae + O2
heat →
""
```

These reactions describe how the shrimps consume algae and oxygen while producing waste and carbon dioxide. The bacteria transform the waste into nutrients, such as nitrates and phosphates, that act as fertilizer for the algae. These nutrients together with carbon dioxide allow the algae to grow, while using light for photosynthesis and producing oxygen as a byproduct. The additional algae feed the shrimps, so that these can survive and sustain the initial reaction.

This network of reactions includes three agencies: shrimps, algae, and bacteria. Their actions are perfectly complementary: each produces what the others consume and consumes what the others produce. The only resource that needs to come from outside the system to keep it going is light. Its only output is heat. That is because such a self-maintaining system, even when it recycles most of the products it uses, must be thermodynamically open in order not to degrade to its maximum entropy state in which everything decays.

It may seem strange that the amount of resources produced and consumed would remain perfectly balanced, so that the system maintains itself over the years. The explanation is that the different agencies will adjust their production and consumption of resources until they are mutually adapted. Suppose that initially there are too many algae for the amount of nutrients available. More algae means more food for the shrimps and therefore more waste and thus nutrients produced. Thus, the amount of algae will diminish by increased consumption, while the amount of nutrients and algae will increase, until both are in balance. Suppose now

that there are too few algae initially. This may lead to the dying off of some of the shrimps. The reduced consumption of algae feeds the shrimps into nutrients. Fewer shrimps means less consumption of algae. This together with the additional nutrients will allow the algae to multiply, until their number is in balance with that of the other components of the system.

Such a closed ecosystem illustrates a very important concept in reaction networks: a (chemical) organization. This concept was introduced by Peter Dittrich, thus founding an approach known as Chemical Organization Theory (Dittrich and Fenizio 2007; Heylighen, Beigi, and Veloz 2015; Veloz et al. 2022). In this theory, an organization is defined as a network of reactions and resources (also called “molecules” or “species”) that is closed and self-maintaining. Closed here means that no resources are produced by the reactions that are not already part of the network. Self-maintaining means that all resources that are consumed by some reactions are produced at least as much by other reactions, so that their total amount does not go to zero. In other words, the resources in a chemical organization are fully recycled and thus remain perpetually available. Interestingly, it can be shown that the attractors of the dynamical system defined by the network of reactions are all chemical organizations (Peter and Dittrich 2011). That means that the system tends to spontaneously settle in one of these organizations, and that once there, it will remain there. In other words, such self-maintaining, closed networks tend to self-organize. Self-organization, as observed long ago by the cybernetician Ashby, can be conceptualized as the mutual adaptation or alignment of the different subsystems within an overall system (Ashby 1962; Heylighen 2001).

...

Indeed, from a relational agency perspective, a living cell is not essentially different from an ecosphere. Both are to some degree transparent containers, filled with a diversity of interacting “molecules” or “species” that are continually being consumed and produced, together forming a self-maintaining, closed network of reactions. Thus, the overall system is **autopoietic**, i.e., self-producing (Heylighen, Beigi, and Veloz 2015; Mingers 1994; Razeto-Barry 2012). The complex, branching cycles of consumption and production reactions constitute the cell’s **metabolism**. While we as yet do not know all the reactions in a bacterial cell’s metabolism, those parts we do know, such as *E. coli*’s sugar metabolism, are easily modeled as chemical organizations (Centler et al. 2007). In the cell, the role of the **resources** is played by “passive” molecules, such as glucose, ATP, and oxygen, which are consumed and produced by reactions in order to harness energy or build components. The role of the **agencies** is played by the **enzymes**, which catalyze and thus enable most of these reactions. Yet, the enzymes themselves are the product of more complex **gene-expressing processes**, which read a coding sequence of DNA and translate it into the right enzyme. As long as the cell does not divide, the DNA itself is **invariant**, i.e., neither

produced nor consumed. That allows it to function as a dependable memory for the cell, storing the “knowledge” about which actions (enzyme-catalyzed reactions) to perform in order to deal with various conditions (presence or absence of particular molecules in the cell).

...

From this perspective, the genes in a cell are functionally similar to the different agencies in a self-maintaining ecosystem. The reactions they trigger must be coordinated or mutually adapted, so that the one produces what the others consume, and the system as a whole remains in balance—while being quick to adjust to fluctuations in the concentrations of the different molecular “species”, or to the entry of food molecules that must be consumed or toxins that must be neutralized. That means that for a cell to survive, its genes must efficiently cooperate, forming a synergetic whole where everything needed by one process is produced by one or more other processes, and vice versa. Note that these processes generally include reactions that transport resources into the cell, and waste out of the cell. Therefore, the system is not closed in the thermodynamic sense, only in the organizational sense.

...

What the cell example further adds to the one of a self-maintaining ecosystem is that the cellular organization produces its own boundary, the cell membrane. This membrane is semi-permeable: it allows “food” molecules to enter, and “waste” molecules to exit, but otherwise protects the self-producing metabolism from external molecules that may interfere with its reactions. That makes the autopoietic organization more robust or resilient (Veloz et al. 2022), and therefore less dependent on outside conditions, i.e. more autonomous. Its implicit goal is self-maintenance (and eventual replication). Therefore, we can view the cell as a whole as an agency in its own right, a “super-agency” consisting of genes and enzymes as “sub-agencies” that control its internal metabolism. This agency acts on its environment by transforming certain available inputs into outputs that change the environmental situation. If we use a unicellular alga as an example, we are back to the reaction that started this series of examples: $\text{alga} + \text{light} + \text{CO}_2 + \text{nutrients} \rightarrow 2 \text{ alga} + \text{O}_2$ This shows that the relational agency ontology, with its formalization in terms of reaction networks and Chemical Organization Theory, is applicable at all levels of life, from molecules to the planetary ecosystem